

咖啡烘焙技能

实操考试 (学员版)

Roasting Professional

Practical Exam

共四大题 SECTION	考试时间 (分钟) MAXIMUM TIME (MINS)	总分 POINTS AVAILABLE
1. 咖啡烘焙 ROASTING	150	48
2. 目测咖啡烘焙度 VISUAL IDENTIFICATION OF ROAST COLOUR	10	12
3. 辨别烘焙瑕疵 IDENTIFICATION OF ROAST DEFECTS	20	30
4. 烘焙品质控制 ROAST QUALITY ASSURANCE	20	20
总计 TOTAL	200 MINS	110

第 1 大题 – 咖啡烘焙 Section 1 – Roasting

考试时间：150 分钟 Time Available: 150 Minutes

总分：48 分 Points Available: 48 points

- 每人 1 台烘焙机。
1 person per roaster.
- 本大题乃时间 x 温度实操测验，可使用任何品牌的烘焙记录仪软件进行记录。
This is a practical time x temp test, and it is to be logged using roast logger software of any brand.
- 本大题旨在烘豆至指定颜色，并使烘焙曲线限制在较小的范围内，以展现您有能力控制烘焙过程，达到各种极限。
The goal is to reach the target color and make the profile follow a path within a narrow limit to show that you are able to control a roast to different extremes.
- 考试期间不得使用曲线记录软件中的参考曲线。
No reference profile can be used on the profile logging software throughout the examination process.
- 根据以下规则，您于 150 分钟内最多可以进行 6 次烘焙。请准备一定量的咖啡生豆，以便在规定时间内完成连续烘焙。
During the 150 minutes, you could end up roasting a maximum of 6 roasts according to the rules below. Prepare an amount of green coffee so that you can complete back-to-back roasts during the allotted time.
- 您可以任意读取咖啡生豆的资料，但限于规定时间内进行。
You may take any readings you wish of the green coffee and this must take place within the allocated time.

1.1 烘焙 1 - 预热和确定最大 ROR

1.1 ROAST 1 - PREHEATING AND MAXIMUM-ROR DISCOVERY

1. 将烘焙机预热到讲师指定的温度。
Preheat the roaster to a temperature specified by the trainer.
2. 用全火 (或讲师指定的最大火力) 进行烘焙 , 直到烘焙至第一次爆裂和第二次爆裂间的中间点。
Perform a roast with full flame (or the max flame provided by the trainer) all the way to the midpoint between 1st and 2nd crack.
3. 在烘焙记录仪软件上记录整个烘焙过程。烘焙过程中不得使用背景参考曲线。
Log the whole roasting process on the roast logger software. No background reference profile is allowed during the roast.
4. 烘焙过程中的最大升温率 (每 30 秒) 为 : _____。
The maximum rate of rise (per 30 seconds) during this roast is: _____.

1.2 烘焙 2 - 参考色值制作 (10 分)

1.2 ROAST 2 - REFERENCE COLOR CREATION (10 POINTS)

1. 将烘焙机预热到讲师指定的温度。
Preheat the roaster to a temperature specified by the trainer.
2. 根据烘焙 1 的结果 , 规划一个受控的烘焙曲线 , 通过烘焙过程调整燃烧器 , 以达到以下参考烘焙指标 :
Based on the result of Roast 1, plan a controlled roast profile where the burner is adjusted through the roast cycle to achieve the following reference roast specification:
 - 最终颜色介于 Agtron (或同等色度计) 75-95 之间
An end color between Agtron (or equivalent colour meter) 75 – 95
 - 第一次爆裂必须发生在以下范围内 : 8:30 分钟至 11:00 分钟
First crack must be in the range: 8:30 minutes to 11:00 minutes
 - 发展时间必须在以下范围内 : 1:45 分钟至 2:45 分钟
Development time must be in the range: 1:45 minutes to 2:45 minutes
 - 咖啡豆测温热电偶的上升率 (ROR) 必须始终高于每 30 秒 0.1°C/1°F。
Rate of rise for the bean temperature thermocouple must always be above 0.1°C/1°F per 30 seconds.
3. 使用烘焙记录仪软件打印烘焙记录 , 签字后交给 AST。
Print the roast from the roast logger software, sign it and hand it to the AST.
4. 从烘焙机中取出咖啡豆并置于室温时 , AST 将记录此次烘焙的咖啡粉色值 (基于 3 个读数的平均值) 。这将成为烘焙 3 和 4 的参考色值。
When the beans are taken out of the roaster and are at room temperature, the AST will record the ground color measurement of this roast (based on an average of 3 readings). This will become your reference color for roasts 3 and 4.
5. 参考色值为 The reference color is : _____。

1.3 烘焙 3 - 快节奏烘焙 (19 分) ROAST 3 - FAST DEVELOPMENT (19 POINTS)

1. 准备一份烘焙 2 的样本，以便在烘焙过程中进行颜色比对。
Have a sample of Roast 2 available to color match during this roast.
2. 规划一个受控的烘焙曲线，通过烘焙过程调整燃烧器，以达到以下指标：
Plan a controlled roast profile where the burner is adjusted through the roast cycle to achieve the following specification:
 - 最终颜色与记录中的烘焙 2 颜色匹配 (+/- 3 Agtron 值)
An end color matching the recorded color of Roast 2 (+/- 3 Agtron points)
 - 第一次爆裂必须发生在以下范围内：7:30 分钟至 10:00 分钟
First crack must be in the range: 7:30 minutes to 10:00 minutes
 - 发展时间必须为 1:30 分钟 (+/-20 秒)
Development time must be 1:30 minutes (+/- 20 seconds)
 - 在烘焙过程中的某个时刻，咖啡豆测温热电偶的上升率应等于或超过烘焙 2 最大 RoR。
Rate of rise for the bean temperature thermocouple should, at some point during the roast, be equal to or exceed the maximum RoR on Roast 2.
 - 咖啡豆测温热电偶的上升率必须始终高于每 30 秒 0.1°C/1°F。
Rate of rise for the bean temperature thermocouple must always be above 0.1°C/1°F per 30 seconds.
3. 使用烘焙记录仪软件记录烘焙过程，打印并交给 AST。
Record the roast in the roast logger software, print it and hand it to the AST.
4. AST 将记录烘焙的咖啡粉色值。
The AST will record the ground color for the roast.
5. 如果以下评分中列出的任何要素需要重新烘焙（限一次），将由 AST 通知您。
If any element outlined in the scoring below requires a re-roast (one permitted) the AST will notify you.

1.4 烘焙 4 – 慢节奏烘焙（19 分） ROAST 4 - SLOW DEVELOPMENT (19 POINTS)

1. 规划一个受控的烘焙曲线，通过烘焙周期调整燃烧器，以达到以下指标：
Plan a controlled roast profile where the burner is adjusted through the roast cycle to achieve the following specification:
 - 最终颜色与纪录中的烘焙 2 颜色匹配 (+/- 3 Agtron 值)
An end color matching the recorded color of Roast 2 (+/- 3 Agtron points)
 - 第一次爆裂必须发生在以下范围内：9:30 分钟至 12:00 分钟
First crack must be in the range: 9:30 minutes to 12:00 minutes
 - 发展时间必须为 3:00 分钟 (+/-20 秒)
Development time must be 3:00 minutes (+/- 20 seconds)
 - 咖啡豆测温热电偶的升温率从未等于或高于烘焙 2 最大 RoR。
Rate of rise for the bean temperature thermocouple is never equal to or higher than the maximum RoR on Roast 2.
 - 咖啡豆测温热电偶的上升率必须始终高于每 30 秒 0.1°C/1°F。
Rate of rise for the bean temperature thermocouple must always be above 0.1°C/1°F per 30 seconds.
2. 使用烘焙记录仪软件记录烘焙过程，打印并交给 AST。
Record the roast in the roast logger software, print it and hand it to the AST.
3. AST 将记录烘焙的咖啡粉色值。
The AST will record the ground color for the roast.
4. 如果以下评分中列出的任何要素需要重新烘焙（限一次），将由 AST 通知您。
If any element outlined in the scoring below requires a re-roast (one permitted) the AST will notify you.

AST 评分 AST SCORING

烘焙方式 Roast Type	烘焙次数 Attempts Roast	总时间 Total Time	一爆时间 Time of 1 st Crack	发展时间 Development Time	最大 RoR Max RoR	最小 RoR Min RoR	颜色 Color
快速 Fast							
评分 Score							
第 1 大题第 2 小题总分 (共 10 分) TOTAL SCORE SECTION 1.2 (10 points available)							

烘焙方式 Roast Type	烘焙次数 Attempts Roast	总时间 Total Time	一爆时间 Time of 1 st Crack Crack	发展时间 Development Time	最大 RoR Max RoR	最小 RoR Min RoR	颜色 Color
参考 Reference							
评分 Score							
第 1 大题第 3 小题总分 (共 19 分) TOTAL SCORE SECTION 1.3 (19 points available)							

烘焙方式 Roast Type	@烘焙次数 Attempts @ Roast	总时间 Total Time	一爆时间 Time of 1 st Crack	发展时间 Development Time	最大 RoR Max RoR	最小 RoR Min RoR	颜色 Color
慢速 Slow							
评分 Score							
第 1 大题第 4 小题总分 (共 19 分) TOTAL SCORE SECTION 1.4 (19 points available)							

总分 (共 48 分)	
TOTAL SCORE (48 points available)	

第 2 大题 - 目测咖啡烘焙度

SECTION 2 – VISUAL IDENTIFICATION OF ROAST COLOR

考试时间：10 分钟 Time Available: 10 Minutes

总分：12 分 Points Available: 12 points

- 考试应保持安静。Silent working is required.
- 色标采用 Agtron 色度分级。The color scale used is the Agtron Gourmet Scale.
- 如果您习惯使用其他商用量测仪进行烘焙颜色测量，AST 将提供换算表。The AST will provide you with conversion tables if you normally use other commercial scales for roast color measurement.
- 本测试旨在评估您能否以目视方式对烘焙颜色进行评分。This test is designed to evaluate your ability to score a roast color using your own eyes only.

1. 您面前有 5 份由 AST 准备的样本，均介于 50 至 90 Agtron 范围内。您有 5 分钟的时间对这些样本进行定标。

In front of you are 5 samples in the range of 50 to 90 Agtron prepared by your AST. You have 5 minutes to calibrate yourself to these samples.

2. 您面前的考试用豆已由 AST 测定，您的任务是为 3 款咖啡豆打分，分值不得超过 AST 测定值 4 度。
The exam coffees in front of you have been measured by the AST, and your task is to score the 3 coffees within 4 points of the AST's measured values.

3. 计分标准 You will receive:

- 4 分 = 估计值超出范围 0–4 度
4 points = estimate is 0 – 4 points out of range
- 3 分 = 估计值超出范围 4.1–8 度
3 points = estimate is 4.1 – 8 points out of range
- 2 分 = 估计值超出范围 8.1–12 度
2 points = estimate is 8.1 – 12 points out of range
- 1 分 = 估计值超出范围 12.1–16 度
1 point = estimate is 12.1 – 16 points out of range

咖啡 1 COFFEE 1		得分 SCORE
您的色值估计： Your color estimate:		

咖啡 2 COFFEE 2		得分 SCORE
您的色值估计： Your color estimate:		

咖啡 3 COFFEE 3		得分 SCORE
您的色值估计： Your color estimate:		

总分 (共 12 分) TOTAL SCORE (12 points available)		
--	--	--

第 3 大题 - 辨别烘焙瑕疵

SECTION 3 – IDENTIFICATION OF ROAST DEFECTS

考试时间：20 分钟 Time Available: 20 Minutes

总分：30 分 Points Available: 30 points

- 考试应保持安静 Silent working is required.
- 必须在红光条件下进行测试。Test must take place under red light conditions.
- 采用三角杯测法，即三种咖啡豆为一组，其中一种咖啡豆与其他两种不同。This is a triangulation test where one coffee within a set of three is different from the other two.
- 本测试旨在评估您能否通过咖啡杯测，辨别常见的烘焙瑕疵并描述其特征。This test is designed to evaluate your ability to identify common roast defects through a cupping exercise and describe the characteristics of the defect.

1. 您将看到以三角杯测形式编码的 6 组杯子（3 个一组）。

You will be presented with 6 sets of 3 cups coded in triangulation test format.

2. 品尝三个样本，圈选每组中异常样本的代码（2 分）。

Taste the three samples and circle the code of the odd sample in each set (2 points).

然后 Then

3. 确定异常样本是目标豆还是瑕疵豆（1 分）。

Identify if the odd sample is a target roast or a defect roast (1 point).

4. 如果是瑕疵，请说出瑕疵类型：浅焙、深焙、焙烤、灼烧或发展不足（1 分）。

If it is a defect, name the defect type: light, dark, baked, scorched or underdeveloped (1 point).

5. 描述该烘焙类型的一项主要感官特征（1 分）。

Describe one dominant sensory characteristic of the roast type (1 point)

三角杯测 1 TRIANGLE 1				得分 SCORE
圈选异常样本的代码： Circle the code of the odd sample:	771	803	287	
是否为烘焙瑕疵？ Is this a roast defect?				
如果是，属何种瑕疵？ If yes, what is the defect?				
请说出这种瑕疵的主要感官特征 Name a dominant sensory characteristic of this defect				

三角杯测 2 TRIANGLE 2				得分 SCORE
圈选异常样本的代码： Circle the code of the odd sample:	781	191	653	
是否为烘焙瑕疵？ Is this a roast defect?				
如果是，属何种瑕疵？ If yes, what is the defect?				
请说出这种瑕疵的主要感官特征 Name a dominant sensory characteristic of this defect				

三角杯测 3 TRIANGLE 3				得分 SCORE
圈选异常样本的代码： Circle the code of the odd sample:	324	673	449	
是否为烘焙瑕疵？ Is this a roast defect?				
如果是，属何种瑕疵？ If yes, what is the defect?				
请说出这种瑕疵的主要感官特征 Name a dominant sensory characteristic of this defect				

三角杯测 4 TRIANGLE 4				得分 SCORE
圈选异常样本的代码： Circle the code of the odd sample:	989	055	851	
是否为烘焙瑕疵？ Is this a roast defect?				
如果是，属何种瑕疵？ If yes, what is the defect?				
请说出这种瑕疵的主要感官特征 Name a dominant sensory characteristic of this defect				

三角杯测 5 TRIANGLE 5				得分 SCORE
圈选异常样本的代码： Circle the code of the odd sample:	773 773	547 547	278 278	
是否为烘焙瑕疵？ Is this a roast defect?				
如果是，属何种瑕疵？ If yes, what is the defect?				
请说出这种瑕疵的主要感官特征 Name a dominant sensory characteristic of this defect				

三角杯测 6 TRIANGLE 6				得分 SCORE
圈选异常样本的代码： Circle the code of the odd sample:	002	195	865	
是否为烘焙瑕疵？ Is this a roast defect?				
如果是，属何种瑕疵？ If yes, what is the defect?				
请说出这种瑕疵的主要感官特征 Name a dominant sensory characteristic of this defect				

总分（共 30 分） TOTAL SCORE (30 points available)				
---	--	--	--	--

第 4 大题 - 烘焙品质控制

SECTION 4 – ROAST QUALITY ASSURANCE

考试时间：20 分钟 Time Available: 20 Minutes

总分：20 分 Points Available: 20 points

- 考试应保持安静。Silent working is required.
- 必须在红光条件下进行测试。Test must take place under red light conditions.
- 本测试旨在评估您能否分辨烘焙曲线的大小差异，模拟生产质量控制流程。This test is designed to evaluate your ability to distinguish between small and large differences in a roast profile to simulate production quality control processes.
- 您必须对生产过程中的变化进行评估，并判断这些变化是否符合规范。You have to assess variation in your production process and decide if the variations are IN or OUT of specification.

1. 您面前有 5 杯熟豆。其中一杯为参考组，供您与自行烘焙的咖啡豆进行比较。

In front of you are 5 coffees. One of them is the reference against which you will compare your production roast.

2. 在 4 杯熟豆中，每种烘焙都属于以下质量控制类别之一：

Amongst the 4 production roasts, each roast falls under one of the following quality control categories:

- 类别 1：符合标准

Category 1: On Standard

- 类别 2：最低标准（即：略有不同，但对于不熟悉咖啡的客户而言，差异并不明显，因此可提供给大多数客群）

Category 2: Bottom Standard (i.e.:is slightly different but not noticeably different by an uneducated customer so it can be released to most customer segments)

- 类别 3：略超出规范（即：对于熟悉咖啡的客户而言，与标准相差太大，但对于特定目标客群而言，仍在其容许范围内而愿意接受）

Category 3: Just Out of Specification (i.e.:is too different from standard for an educated customer but could be released depending on the variation tolerance threshold for the specific customer segment targeted)

- 类别 4：不符合规范（即：差异太大，无法向任何客群提供）

Category 4: Out of Specification (i.e.:is too different to be released to any customer segment)

3. 每个样本的评分标准如下：

The scoring for each sample is as follows:

- 辨别瑕疵 = 1 分

Identify defect =1 point

- 说出瑕疵名称 = 1 分

Name roast defect = 1 point

- 分类结果 = 3 分

Category outcome =3 points

代码 CODE	瑕疵 (是/否) DEFECT (Y/N) 1 分 1 point		烘焙瑕疵 1 分 ROAST DEFECT 1 point		分类结果 3 分 CATEGORY OUTCOME 3 points				
497	是 Yes		无瑕疵 No Defect		1 符合标准 - 及格 On Standard - Pass				
			极浅焙 Very Light						
			浅 Light		2 微小差异，符合最低标准 - 及格 Small Difference Bottom Standard - Pass				
			深焙 Dark						
	否 No		极深焙 Very Dark		3 略超出规范 - 不及格 Just Out of Specification – Fail				
			焙烤 Baked						
								总分 (共 5 分) TOTAL SCORE (5 points available)	

代码 CODE	瑕疵 (是/否) DEFECT (Y/N) 1 分 1 point		烘焙瑕疵 1 分 ROAST DEFECT 1 point		分类结果 3 分 CATEGORY OUTCOME 3 points	
423	是 Yes		无瑕疵 No Defect		1 符合标准 - 及格 On Standard - Pass	
			极浅焙 Very Light			
			浅 Light		2 微小差异，符合最低标准 - 及格 Small Difference Bottom Standard - Pass	
			深焙 Dark			
	极深焙 Very Dark		4 明显超出规范 - 不及格 Noticeably Out of Specification - Fail			
	焙烤 Baked					
	否 No					
总分 (共 5 分) TOTAL SCORE (5 points available)						

代码 CODE	瑕疵 (是/否) DEFECT (Y/N) 1 分 1 point		烘焙瑕疵 1 分 ROAST DEFECT 1 point		分类结果 3 分 CATEGORY OUTCOME 3 points	
441	是 Yes		无瑕疵 No Defect		1 符合标准 - 及格 On Standard - Pass	
			极浅焙 Very Light			
			浅 Light		2 微小差异 · 符合最低标准 - 及格 Small Difference Bottom Standard - Pass	
	否 No		深焙 Dark		3 略超出规范 - 不及格 Just Out of Specification – Fail	
			极深焙 Very Dark			
			焙烤 Baked		4 明显超出规范 - 不及格 Noticeably Out of Specification - Fail	
	总分 (共 5 分) TOTAL SCORE (5 points available)					

代码 CODE	瑕疵 (是/否) DEFECT (Y/N) 1 分 1 point		烘焙瑕疵 1 分 ROAST DEFECT 1 point		分类结果 3 分 CATEGORY OUTCOME 3 points	
580	是 Yes		无瑕疵 No Defect		1 符合标准 - 及格 On Standard - Pass	
			极浅焙 Very Light			
			浅 Light		2 微小差异 · 符合最低标准 - 及格 Small Difference Bottom Standard - Pass	
	否 No		深焙 Dark		3 略超出规范 - 不及格 Just Out of Specification – Fail	
			极深焙 Very Dark			
			焙烤 Baked		4 明显超出规范 - 不及格 Noticeably Out of Specification - Fail	
	总分 (共 5 分) TOTAL SCORE (5 points available)					

总分 (共 20 分) TOTAL SCORE (20 points available)						
--	--	--	--	--	--	--

考试结束 END OF EXAM